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***Abstract*—In this report, we present a regression model capable of estimating a person's age based on an audio recording of his/her speech and associated features. Multiple features have been extracted from the audio recordings available and multiple models have been considered to yield a satisfactory result even if further improvement can be still developed.**

I. Problem overview

In order to build a regression model able to estimate the age of a person, we use a set of 2933 recordings for the training phase and 691 recordings for the evaluation phase of the model. We have been given a dataset that comprehends for both the training and testing set:

- a `csv` file (respectively `development.csv` and `evaluation.csv`) containing some relevant features about the audios;
- the audio recordings, from which other relevant features have been extracted.

For the training set, it is accessible also the data about the *age*, that ranges between 6 and 97. The values are not distributed equally because 75% of people has less than 32 years, as shown in Figure 1.

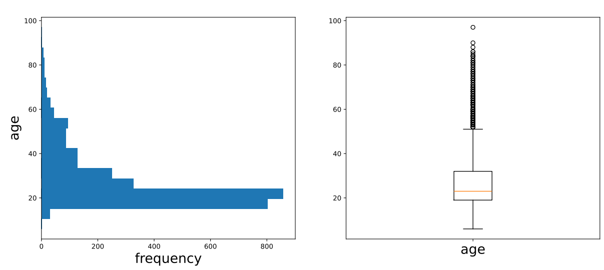

Fig. 1. Distribution of the age in the training set.

The other columns are *sampling_rate, gender, ethnicity, mean_pitch, max_pitch, min_pitch, jitter, shimmer, energy, zcr_mean, spectral_centroid_mean, tempo, hnr, num_words, num_characters, num_pauses, silence_duration, path* .

The first observation when looking into the data is that the sampling rate for the audios is constant for any row, equal to 22050 Hertz. Furthermore, there are no missing values in any of the features and the audios are split equally between male and female speakers.

ethnicity has captured our interest since the number of different ethnicities in the training set is 165, but only 2 of them (*english* and *igbo*) can represent more of 50% of the dataset.

Other features that show an un-ordinary distribution are *num_words* and *num_characters*, shown in Figure 2, which indicate respectively the number of words and characters in a sentence. Since the estimated speaking rate, the number of pauses and duration of silence are more normally distributed, it may indicate different length of the audios and that could be a possible feature for the models.

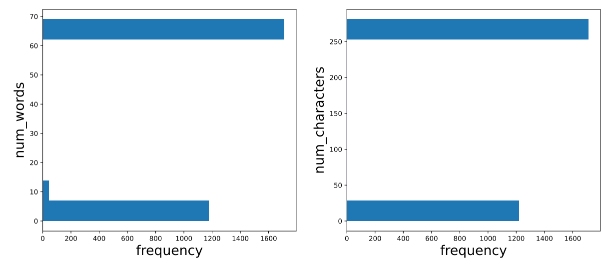

Fig. 2. Distributions of *num_words* and *num_characters*

II. Proposed approach

A. Data preprocessing

The data preprocessing stage will be divided in four steps:

1) De-noising of the audio recordings: Listening to some of the recordings revealed that a part of them present background noise and others have long moments of silence at the start and at the end of the recording. For this reason we thought of performing some data cleaning by trimming and removing the noise, using the functions implemented in *librosa* that preserve only the vocals part of an audio.

We created a cleaned version of all the audios and we will perform a comparison with the originals later.

2) Encoding of not numerical features: All the features given are numerical, except: *path, gender* and *ethnicity*.

- *Path* will be used only for coding purposes so it is irrelevant to the analysis.
- *Gender* has been made binary, with *True* indicating a female speaker and *False* a male one.
- *Ethnicity* was one-hot encoded considering only the first two ethnicity [1].

[1]The first two ethnicities are *English* and *Igbo*, they represent 57% of the training dataset while the third one contributes for only a 3%

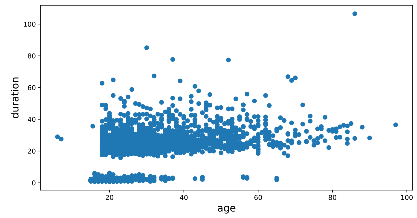

Fig. 3. Scatterplot of *age* with respect of *duration*

3) Feature extraction and selection: New features have been extracted from the audio recordings, as follows:

- *duration*: length in seconds of the recordings. The correlation between it and *age* is 0.532 as seen in Figure 3;
- *temporal features*: the feature extracted by using *maad all temporal* function, without duplicating those already present in the csv.
  - *temporal entropy*;
  - *peak frequency*: frequency with the highest amplitude in the spectrogram;
  - *temporal median*: median of the envelope;
  - *nVoicedFrames*: number of voiced frames;
  - *meanAbsoluteSlop*: changes of the signal between consecutive samples;
  - *energy between 250-600* Hz *and 1-8* kHz
  - *nFrames*: number of frames;
- *fundamental frequency* f_0: in particular some statistics of the fundamental frequency distribution as the mean, the median, the standard deviation, the 5% and 95% percentile;
- *MFCCs*: (Mel Frequency Cepstral Coefficients)
- f_1, f_2, f_3, f_4: the mean and the standard deviation of the first four formants extracted using Burg's algorithm with step step 0.02 and maximum formant 5000 Hz as seen in the literature [1].

In order to reduce the dimensionality and keep only the best features we looked at the covariance with the target value and used a multiple *Random Forest* regressor[2]. An evaluation of the impact of those features has been tested by comparing the results obtained with and without their presence.

Looking at the feature importance of the ethnicity columns, *Igbo* results to be less important than *English*, even if Eboans people are more than the English ones and this may be because of the fluency of English speakers since all the recordings are in English. The contribute of the *Ibgo* column[3] is only 0.000774 while the contribute of the *English* column is 0.047287. Another important observation occurred during the data exploration phase is that the ethnicity *English* is not present in the evaluation set, therefore we decided to drop both these columns.

[2]We prefer a *Random Forest* regressor for this phase because it provides also the possibility to see the feature importance

[3]It has one order in magnitude lower than the second last important feature

Also some of the features extracted have been resulted not really useful, such as the statistics of the fundamental frequency f_0 and therefore they have been discarded.

During the preprocessing phase, the optimal number of MFCCs hasn't been picked up yet. Every coefficient added represents a new feature, hence an extremely large number of MFCCs hasn't been considered in order to avoid the curse of dimensionality. With this naive model, the best value for MFCCs is 13. Using the mean value with respect to the temporal axes, we decided to not use the standard deviation of the MFCCs because the model performance were always worse.

Finally by comparing the performance of the model trained with the feature extracted from the cleaned dataset and those extracted form the original dataset, we found out that the cleaned version performed slightly worse, so we decided to use only the features extracted form the original audios. The worst performance may be caused by an excessive cleaning that leads to losing some information.

B. Model selection

Multiple techniques have been tested for this problem.

Considering the differences between male and female voices, two separate regressors were trained: one exclusively for male recordings and the other for female recordings. The most relevant drawback of this approach is that each model is trained on only half of the available data. However, this trade-off might be justified if a general model struggles to capture the distinctions between male and female voices using only the feature *gender*.

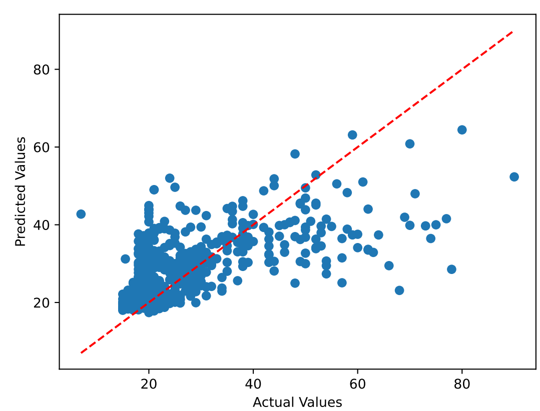

Fig. 4. Predictions of the naive model

As said before, a naive model has been trained for feature selection and looking at its predictions we notice how it overestimates the age of the young people and underestimates the age of older ones. To limit this behaviour we try building a pipeline:

1) a classifier to divide the records into two ranges of ages, i.e. young speakers and old speakers;
2) two regressor where each model is focused on a specific range.

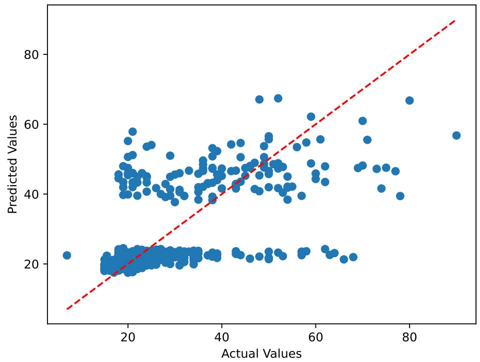

Fig. 5. Prediction of the model build with classifier and two regressors

Finally we tested four other models, that are robust and effective for high-dimensional datasets:

1) Random Forest Regression;
2) Hist Gradient Boosting Regressor;
3) Support Vector Regressor, with a prior standard scaler;
4) Voting Regressor composed by combinations of the three previous regressors.

C. Hyperparameter tuning

After choosing the models, it is necessary also to tune the hyperparameters. In the following Tables I, II, III are presented the chosen hyperparameters for respectively the *Random Forest*, the *Hist Gradient Boosting* and the *SVR* models.

TABLE I
HYPERPARAMETERS FOR RANDOM FOREST

n_estimators	250	500	1000
max_depth	None	10	20
max_features	log2	sqrt	None
min_samples_leaf	1	2	4
min_samples_split	2	4	8

TABLE II
HYPERPARAMETERS FOR HIST GRADIENT BOOSTING

max_iter	100	1000	
learning_rate	0.01	0.1	
l2_regularization	0	0.1	0.01

TABLE III
HYPERPARAMETERS FOR SVR

svr_C	0.1	1	10	100

III. RESULT

All the models have been trained and tested using cross validation with five folds on the development set, with the Root Mean Square Error (RMSE) used as the evaluation metric.

The score obtained from the analysis on the two separate regressors on the two different gender provide a suboptimal performance: the RMSE for the male Random Forest model is $9,63$ while for the female model is 10.13. The inadequate regression obtained and the assumption that a general model could still consider the feature *gender* was ultimately abandoned.

Similarly, the analysis performed on two different ranges of age yielded suboptimal results due to the not great accuracy of the classifier, only 0.84, and the reduced training set for each of the regressors. Indeed the prediction of the model with two classifier were not good as it is visible in Figure 5. Then the poor performances of these two methods reinforced the decision to focus only on a single general model.

TABLE IV
SELECTED HYPERPARAMETERS

Model	**Parameter**	**Best value**	**RMSE**
Random Forest	*n_estimators*	1000	9.64
	max_depth	None	
	max_features	sqrt	
	min_samples_leaf	4	
	min_samples_split	4	
Hist Gradient Boosting	*max_iter*	1000	9.38
	learning_rate	0.01	
	l2_regularization	0.1	
SVR	*C*	10	9.67

This results of the grid searches, visible in Table IV are not exceptional as we would expect.

Then the final step is the *Voting Regressor* on the previous models. The best performance is achieved by the *Voting Regressor* of *Hist Gradient Boosting Regressor* and *SVR* with a test score of 9.23 and an evaluation score of 9.41 in the leaderbord. It is able to perform better as the combined predictions reduce variance, leading to less dispersion in the final output. All the predicted values are shown in Figure 6.

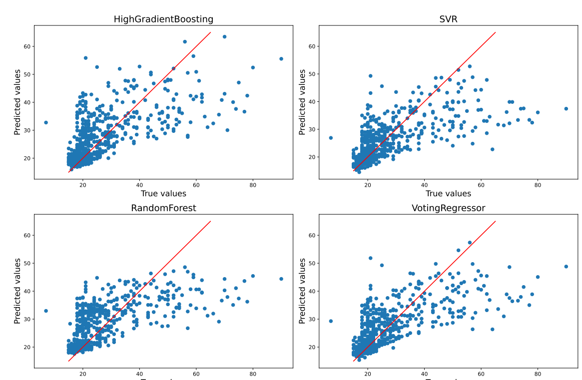

Fig. 6. Prediction vs. Actual values

IV. DISCUSSION

The selected models performs quite good with respect to the baseline RMSE = 13.091[4], but they often do not perform

[4] This is the model that predicts the mean of the observations *age*

enough better than the naive *Random Forest* model. The difficulties arise from the bad distribution of the speakers' age, variability in utterances and differences in the length of the recordings. Indeed a RMSE of 9.41 means that the algorithm gives result with around 9 year of error, in particular the prediction is enough accurate for a young person but less for an older one.

REFERENCES

[1] C. van Heerden, E. Barnard, M. Davel, C. van der Walt, E. van Dyk, M. Feld, and C. Müller, "Combining regression and classification methods for improving automatic speaker age recognition," in *2010 IEEE International Conference on Acoustics, Speech and Signal Processing*, pp. 5174–5177, IEEE, 2010.